RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

MINI PROJECT

DIABETIC PATIENT DATA ANALYSIS AND VISUALIZATION USING POWER BI

Framework for Data Visual Analytics (FDVA)

Submitted by

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Dataset: diabetic_data.csv | Tool: Microsoft Power BI

Abstract

This mini project, carried out for the subject Framework of Data Visualization and Analytics (FDVA), analyses a diabetic patient dataset using Microsoft Power BI. The dataset is stored in diabetic_data.csv and contains 101,766 hospital encounters with 50 columns, such as patient age, gender, race, admission type, admission source, time in hospital, number of lab procedures, number of medications, diagnosis codes, 23 diabetes medication columns, diabetes medication status and readmission. The data was checked for missing values, duplicates and data types. Missing values were stored as the symbol “?” in seven columns, with weight (96.9%), medical specialty (49.1%) and payer code (39.6%) having the most; no duplicate encounters were found. A Power BI report with two pages and six visuals (bar chart, pie charts, donut chart, line chart and area chart) was then built directly on the table.

The analysis showed that patients aged 70–80 years form the largest age group, that 77% of the encounters involved a diabetes medication, that the average stay was 4.4 days, and that lab procedures account for about 63% of the combined procedure, medication and diagnosis counts. The project shows how a few well-chosen visuals can turn a large hospital dataset into clear information, and also how the choice of fields and aggregation affects what a chart can say.

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1. Introduction, Objectives and Tools

Diabetes is a long-term condition that needs repeated hospital care, and hospitals record a large amount of data for every patient visit. Looking at such records row by row is not practical, so visual analytics is used to turn them into charts that show patterns in age, treatment and length of stay at a glance. This project applies that idea to a hospital dataset of patients with diabetes.

Objectives

- To load and inspect the diabetic patient dataset and find missing, duplicate and unusual values.
- To prepare the data so that it can be used correctly in Power BI.
- To build an interactive two-page Power BI report with different chart types.
- To read the main insights from each visual and note where a chart could be improved.

Tools used

- **Microsoft Power BI Desktop** for loading the data, building the visuals and the report pages.
- **diabetic_data.csv** as the single data source, loaded as one table named diabetic_data.
- **Python (pandas, matplotlib)** only to cross-check the counts and to draw the supporting charts in Section 5.

Approach

The work followed four steps: (1) load the CSV file into Power BI, (2) check the data quality, (3) build the visuals using fields directly from the table, and (4) read and note the insights. All numeric fields used the default Sum aggregation in Power BI. The effect of this choice is discussed with each chart.

2. Dataset Description

The dataset (diabetic_data.csv) has 101,766 rows and 50 columns, and each row is one hospital encounter of a patient with diabetes. The 101,766 encounters belong to 71,518 different patients, so many patients appear more than once. Of the 50 columns, 13 are numeric and 37 are text.

Group	Example columns	Type
Identifiers	encounter_id, patient_nbr	Numeric
Patient details	race, gender, age, weight	Text
Admission details	admission_type_id, admission_source_id, payer_code	Code / text
Hospital stay	time_in_hospital, medical_specialty	Numeric / text
Work done	num_lab_procedures, num_procedures, num_medications	Numeric
Visit history	number_outpatient, number_emergency, number_inpatient	Numeric
Diagnosis	diag_1, diag_2, diag_3, number_diagnoses	Text / numeric
Medications (23)	metformin, insulin, tolazamide, tolbutamide ...	Text
Outcome	diabetesMed, readmitted	Text

3. Data Preprocessing

3.1 Missing values and duplicates

Missing values in this dataset are written as the symbol “?” and not as blanks, so Power BI does not count them as nulls. Seven columns contain it. Weight is missing in almost every row and is therefore not suitable for analysis. No duplicate rows were found, and every encounter_id is unique.

Column	Rows with "?"	Share of rows
weight	98,569	96.9%
medical_specialty	49,949	49.1%
payer_code	40,256	39.6%
race	2,273	2.2%
diag_3	1,423	1.4%
diag_2	358	0.4%
diag_1	21	0.02%

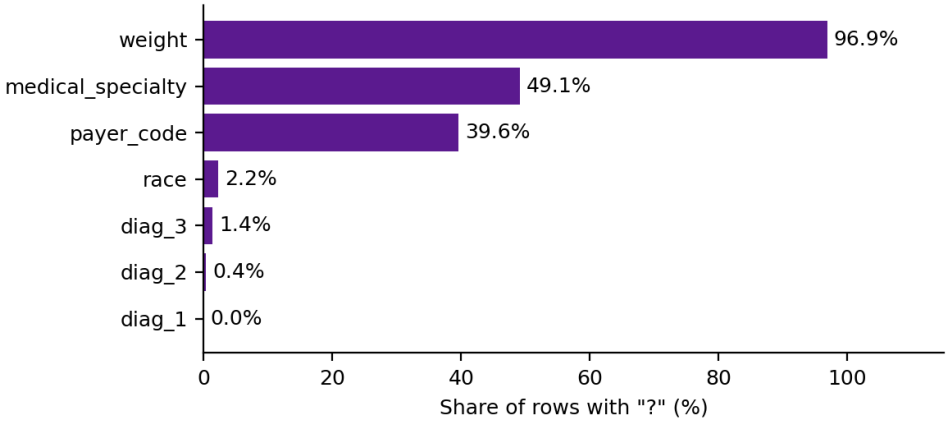


Figure 3.1: Share of missing values (“?”) by column

3.2 Data types and category values

Age is stored as ten-year text bands, from [0-10) to [90-100), and is sorted as text in Power BI, not by age. Gender has 54,708 Female, 47,055 Male and 3 Unknown/Invalid rows. The medication columns (for example insulin, metformin, tolazamide and tolbutamide) hold the values No, Steady, Up or Down, and most of them are almost always No; examide and citoglipton have only one value, so they carry no information. The readmitted column has the values NO (54,864), >30 (35,545) and <30 (11,357), and diabetesMed has Yes (78,363) and No (23,403).

3.3 Numeric columns used in the visuals

The measures num_lab_procedures, num_medications, num_procedures, number_diagnoses, number_emergency, number_inpatient and time_in_hospital are real counts. The columns admission_type_id and admission_source_id, however, are code numbers (for example 1 = Emergency), not quantities. Power BI summed them by default when they were dropped into a chart, and the “?” entries were kept as a category (it appears as the first entry in the diag_2 legend). The effect of this on the charts is explained in the next section.

4. Power BI Dashboard

Power BI was used to build an interactive report with two pages from the diabetic_data table. Every visual uses fields directly from the table, with the default Sum aggregation for numeric fields.

4.1 Page 1

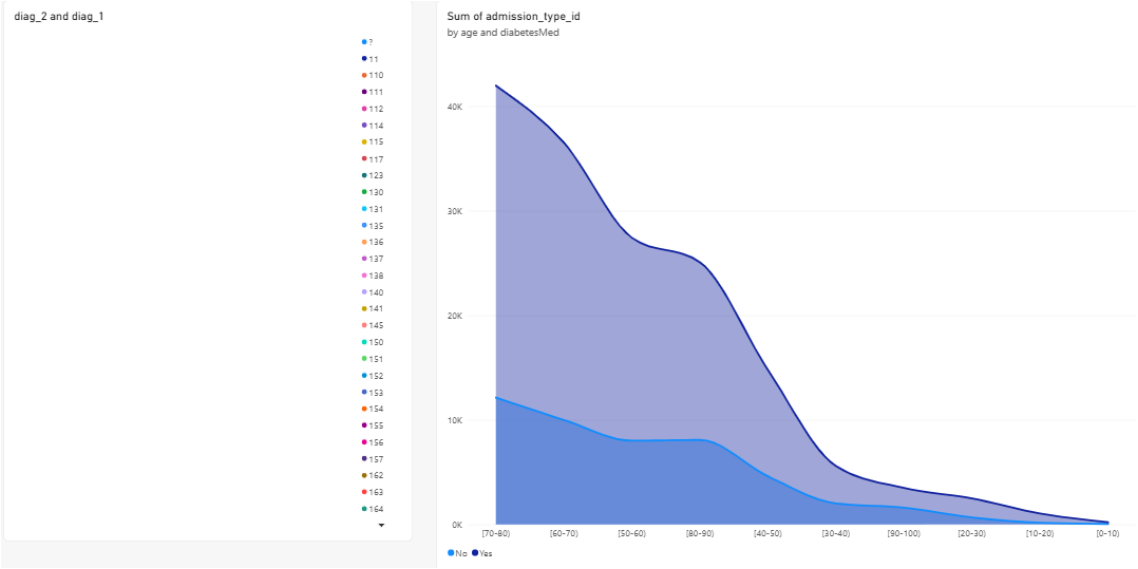


Figure 4.1: Power BI Report – Page 1

- **Charts:** a donut chart of diag_2 with diag_1 as the series, and an area chart of the sum of admission_type_id by age and diabetesMed.
- **Insights:** in the area chart the 70–80 age group is the highest (about 54,000), followed by 60–70 (about 46,500) and 50–60 (about 35,400). The Yes (on diabetes medication) area is above the No area in every age group, which matches the 77% share of Yes in the data. Patients below 30 years are very few.
- **Note:** the donut chart has about 750 diag_2 codes, so its legend is very long and the slices cannot be read. It would be clearer after grouping the codes into disease groups or showing only the top 10. The age axis is sorted by value and not in age order.

4.2 Page 2

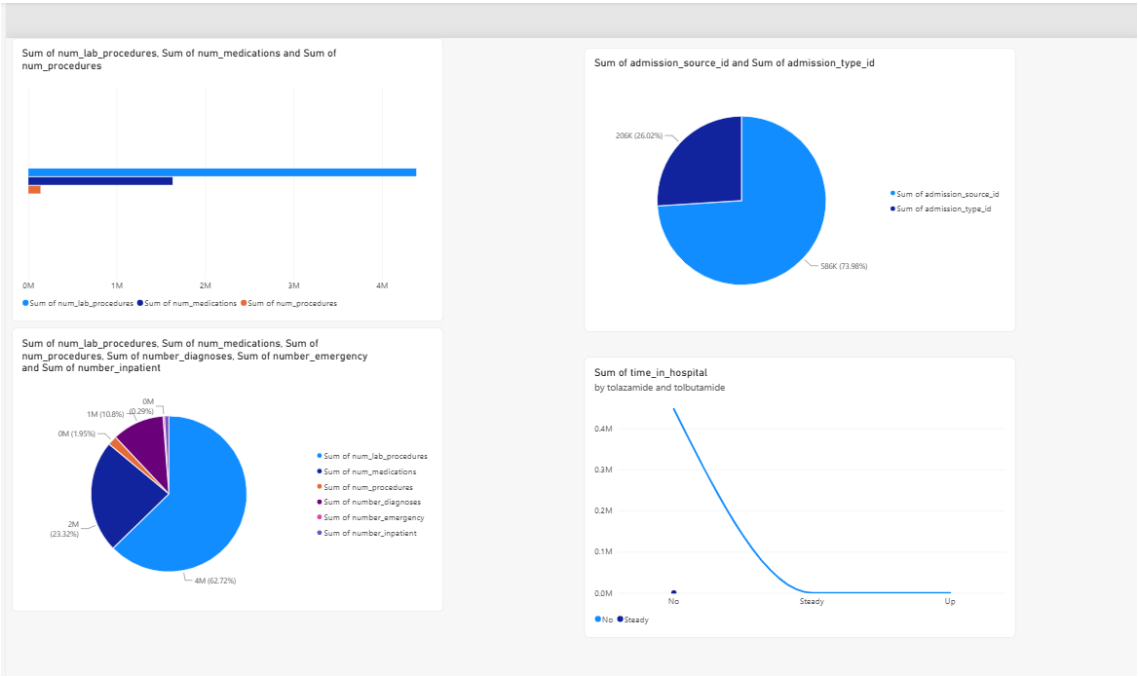


Figure 4.2: Power BI Report – Page 2

- **Charts:** a clustered bar chart and a pie chart of the sums of num_lab_procedures, num_medications and related counts; a pie chart of admission_source_id and admission_type_id; and a line chart of time_in_hospital by tolazamide with tolbutamide as the series.
- **Insights:** the bar chart shows about 4.39 million lab procedures, 1.63 million medications and 0.14 million procedures. In the six-measure pie chart lab procedures take 62.72%, medications 23.32% and diagnoses 10.8%, while procedures (1.95%) and emergency visits (0.29%) are very small. Total time in hospital is about 447,000 days, which is an average stay of 4.4 days per encounter.
- **Note:** the pie chart of admission_source_id and admission_type_id (73.98% and 26.02%) compares sums of code numbers, so it has no real meaning; a count of encounters by admission type would be better. In the line chart almost all stays fall under the value No for both medicines, since tolazamide is Steady in only 38 encounters and tolbutamide in only 23, so the line drops to nearly zero.

Overall, the dashboard shows that the patient group is mostly elderly, that most patients are on diabetes medication and that lab tests are by far the largest part of the work done during a stay.

5. Supporting Analysis of the Dataset

To support the dashboard, the same CSV file was cross-checked in Python. The charts below use counts of encounters, which avoids the problem of summing code numbers seen in Section 4.

5.1 Encounters by age group

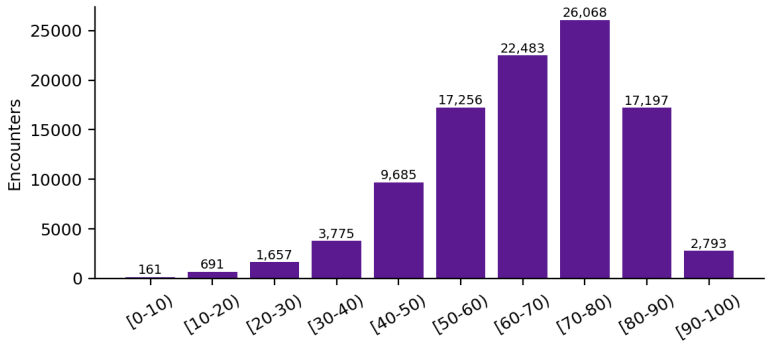
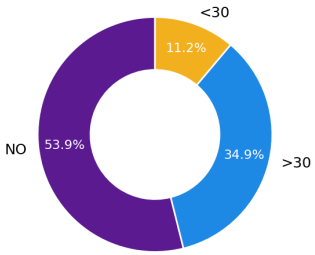


Figure 5.1: Number of encounters by age band (in age order)

Counted properly and sorted in age order, the distribution rises steadily to a peak at 70–80 (26,068 encounters), then falls. The 60–70 band has 22,483 and the 50–60 and 80–90 bands have about 17,200 each. Together the bands from 50 to 90 years hold about 83,000 of the 101,766 encounters.

5.2 Readmission



More than half of the encounters (53.9%) were not followed by a readmission. About 34.9% were readmitted after more than 30 days and 11.2% within 30 days. Readmission within 30 days is the group hospitals usually care about most, as it may point to a stay that was too short or to follow-up care that was not enough.

Figure 5.2: Share of encounters by readmission status

5.3 Readmission by age group

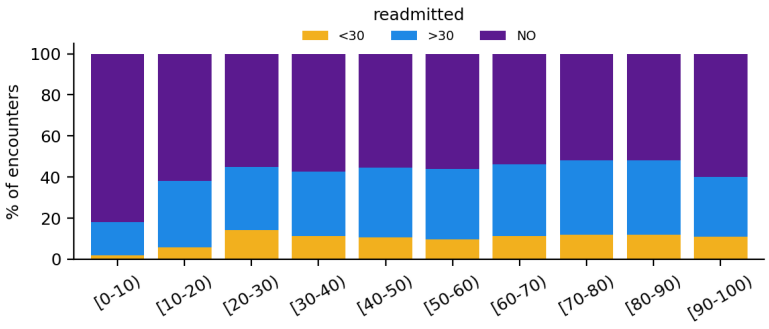


Figure 5.3: Readmission status within each age band

Children under 10 are rarely readmitted (82% have no readmission), while the share with no readmission is lowest, about 52%, in the 70–80 and 80–90 groups. The average stay also grows with readmission risk: 4.25 days for NO, 4.50 days for >30 and 4.77 days for <30.

6. Findings, Limitations and Conclusion

Key findings

- The dataset has 101,766 encounters of 71,518 patients with no duplicates; weight, medical specialty and payer code are largely missing.
- The patient group is mostly elderly: the 70–80 band is the largest, and patients below 30 years are very few.
- About 77% of the encounters involved a diabetes medication (78,363 Yes against 23,403 No).
- The average stay was 4.4 days, with total time in hospital of about 447,000 days.
- Lab procedures make up about 63% of the combined work counts, followed by medications (23%) and diagnoses (11%).
- Only 11.2% of encounters ended in a readmission within 30 days, and older groups have the highest readmission shares.

Limitations and suggested improvements

- **Sums of code columns:** admission_type_id and admission_source_id are codes, so their sums have no meaning. Use a count of encounters by admission type instead.
- **Too many categories:** the diag_2 donut chart has about 750 codes. Group them into disease groups or show the top 10 only.
- **Sparse medication columns:** tolazamide and tolbutamide are almost always No, so the line chart shows almost nothing. Use a drug with more variation, such as insulin or metformin.
- **Age order:** sort the age field by band and not by value, and use Average of time_in_hospital instead of Sum.
- **Missing values:** replace “?” with Unknown or filter it out, and leave out weight.

Conclusion

This mini project used Microsoft Power BI to analyse a hospital dataset of more than 100,000 diabetic patient encounters. After checking the data for missing values, duplicates and data types, a two-page report with six visuals was built. The report showed that the patients are mostly elderly, that most of them are on diabetes medication, that the average stay is about 4.4 days and that lab tests are the largest part of the work done during a stay. The work also showed that a chart is only as useful as the field and aggregation chosen for it: counts and averages on properly grouped fields give clearer and more correct answers than default sums. With these improvements, the report can serve as a simple and useful tool for understanding diabetic patient data.